

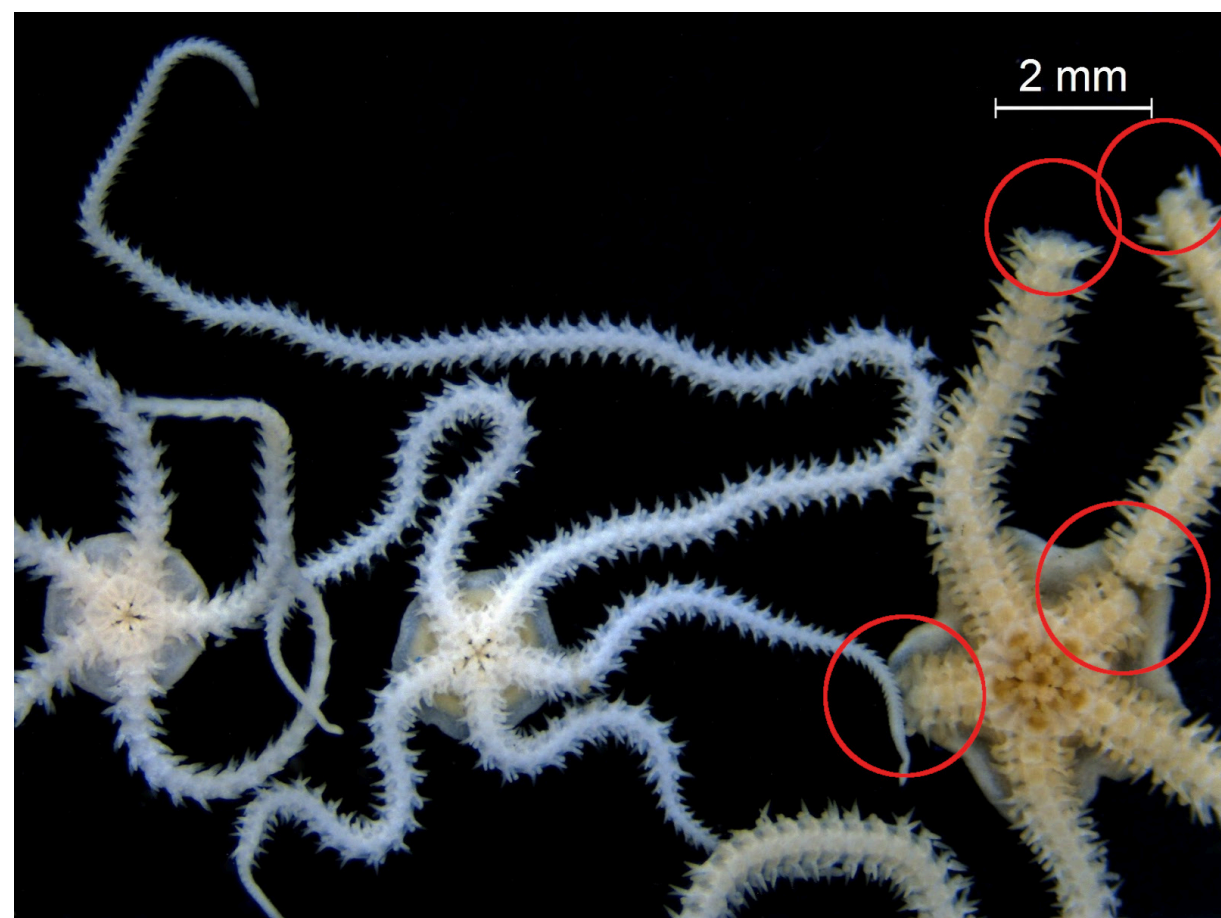
Brittle Star-Inspired Damage Robustness with Multi-Agent Reinforcement Learning

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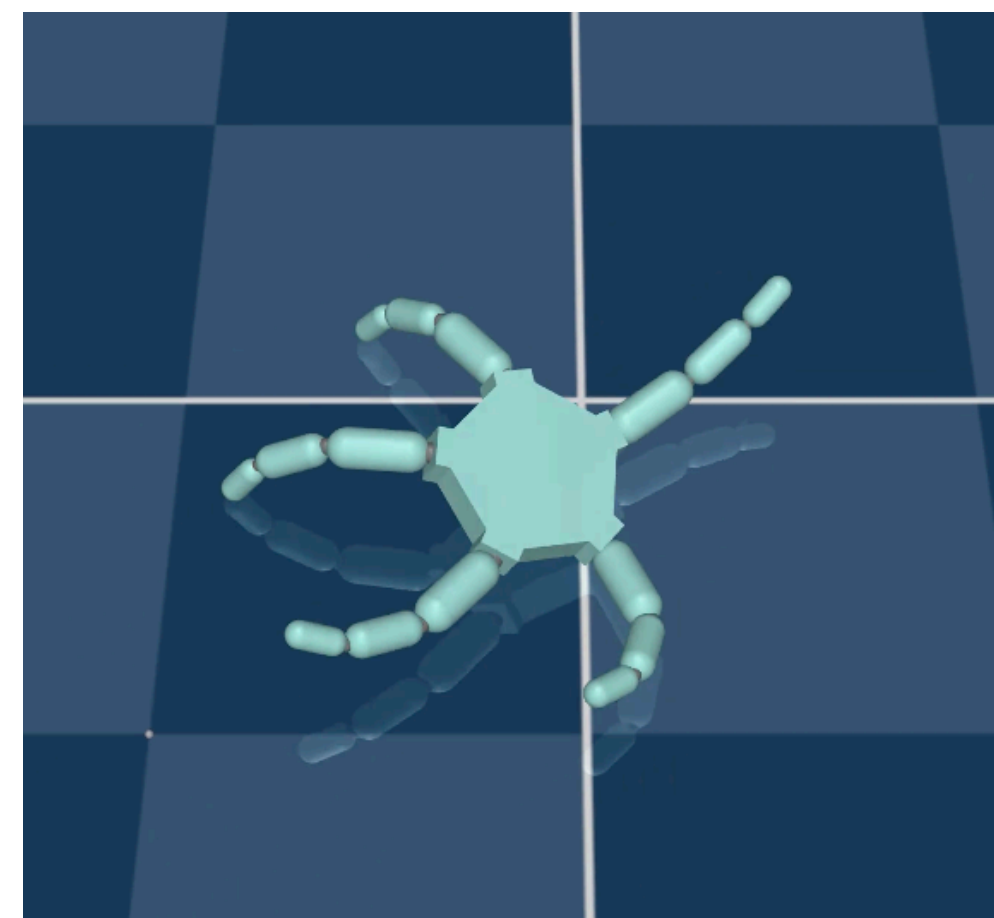
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1 Damage Robustness

Real-world robots might sustain **damage** in operation. Nature already has solutions to this problem, e.g. in the form of **brittle stars**: sea creatures that are close relatives to starfish, but that can deal with the loss of limbs by **adapting their locomotion** and even regeneration.



Two intact and one damaged brittle stars.



Brittle star-inspired robot in a simulated environment.

brittle star damage robustness $\xrightarrow{?}$ damage-robust robots

2 Research Goal

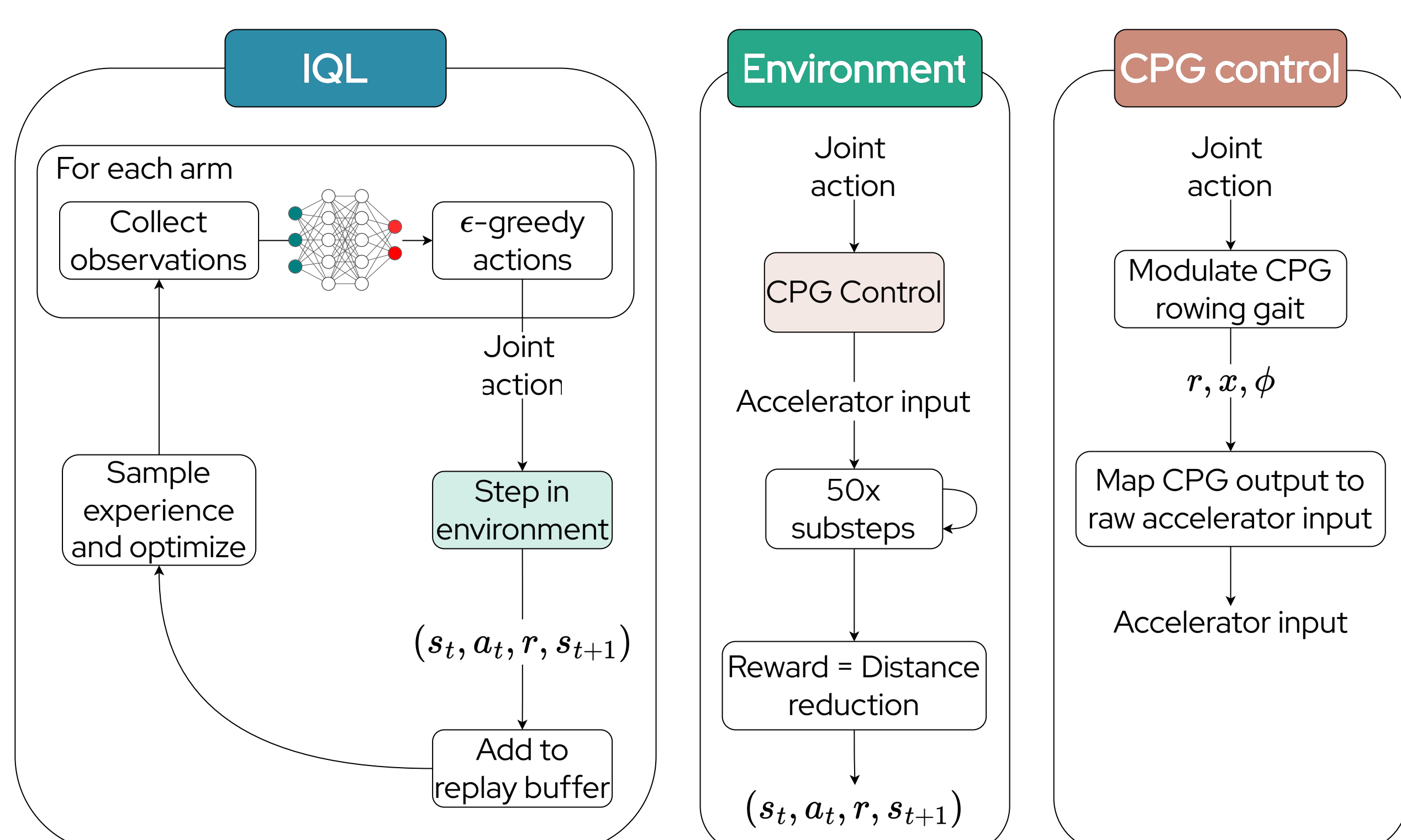
Build damage-robust robots by learning brittle star-inspired locomotion gaits with the help of central pattern generators (CPGs) and multi-agent reinforcement learning (MARL).

3 Methodology

Central Pattern Generators (CPGs) are a **mathematical model for representing natural rhythmic motions** without requiring external steering.

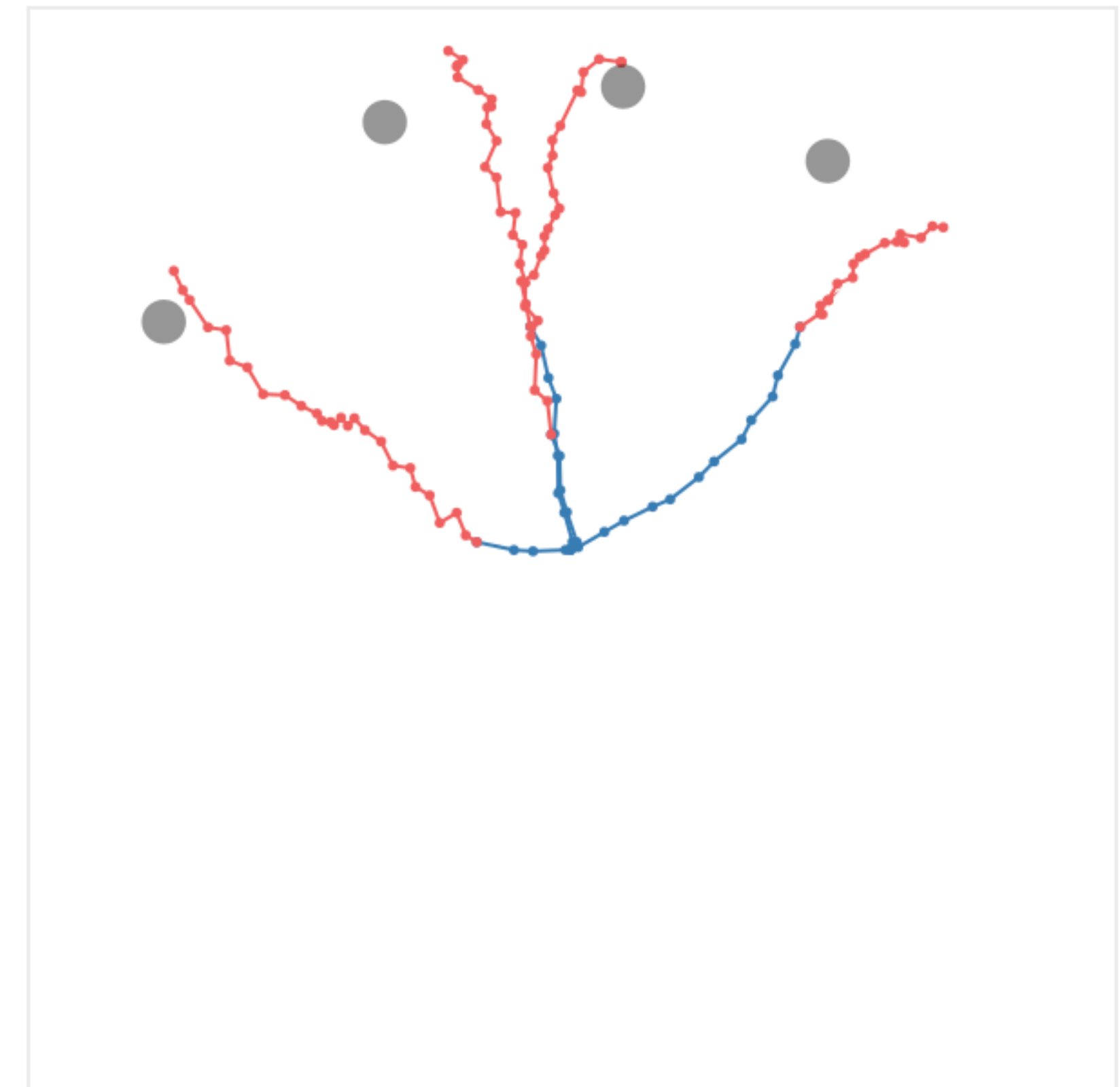
We use them in a **Multi-Agent Reinforcement Learning (MARL)** pipeline where the CPGs are the chosen actions. The idea of MARL is to have **multiple agents interact independently** in the same environment.

We make use of **Independent Q-Learning (IQL)**, which is an adaption of standard Q-Learning in which each agent treats the **other agents as an extra source of stochasticity**. In order to keep training complexity lower, the agents **share their networks' parameters** with each other so they can **share experience** with each other.

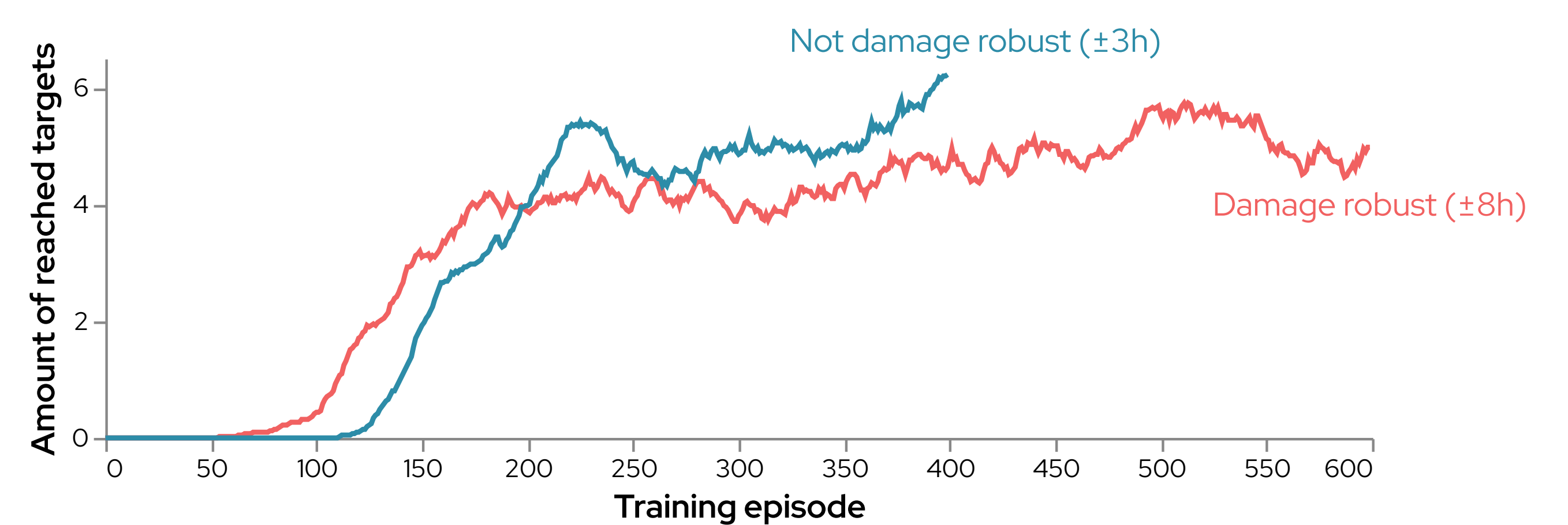


4 Results

Trajectories taken by the Brittle Star
Targets indicated in grey, damaged interval in red



Achieving similar performance with damage robustness
requires more than twice as much time



5 Conclusions

1. Training to walk to a target on a larger distance requires more and longer training episodes.
2. Training with damage takes more than twice as long as training without damage to obtain similar results.

6 Future work

- Explore more complex locomotion gates (without CPGs?)
- Trying out other MARL algorithms